

Drug therapy in Pediatrics

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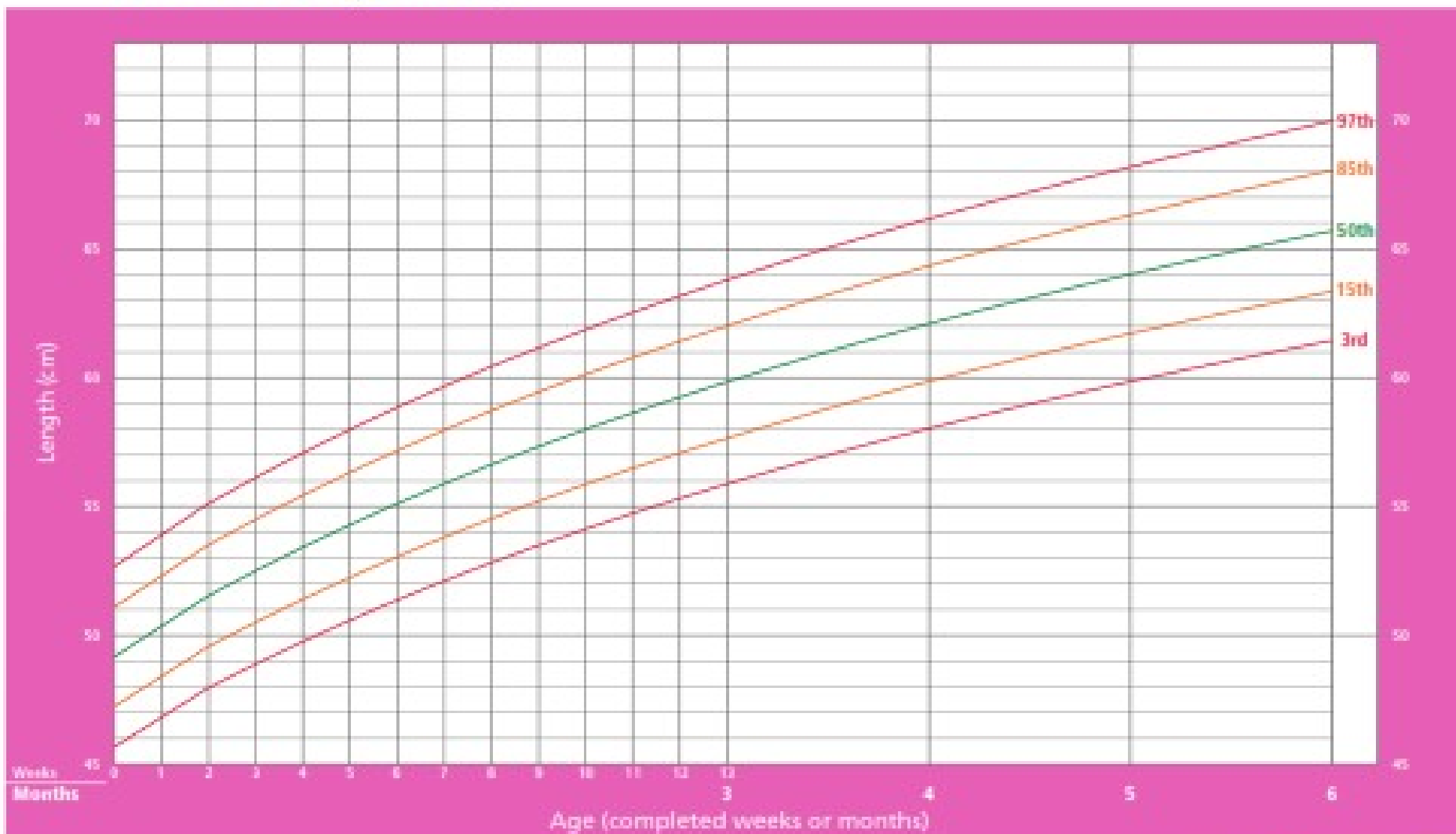
Introduction

- Pediatrics Vs Adults

Length-for-age GIRLS

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Birth to 6 months (percentiles)



Effects of Pharmacokinetic and Pharmacodynamic Differences

- PK/PD may be similar or different depending on the drug.
 - Vancomycin vs. Phenobarbitone
- Determine selection and dosing of drugs
 - Many factors should be considered
- Weight based vs Uniform dose calculation

Pharmacokinetics – Absorption

- Oral – depends on pH, gastric emptying and pancreatic function
- Transdermal – depends on skin structure, perfusion, BSA to weight ratio
- IM – variable absorption in neonates and infants, Painful
- Rectal – erratic absorption; not commonly recommended

Pharmacokinetics – Volume of Distribution

- ECF and TBW higher in neonates and infants
 - Higher V_d for water soluble drugs
- Lower range of plasma proteins
- Adipose composition
- Blood brain barrier

Pharmacokinetics – Metabolism

- Hepatic metabolism
- Liver to body mass ratio

Pharmacokinetics – Elimination

- Nephrogenesis – completed at 36th week of gestation
- Premature infants – slower GFR
- GFR – increases during childhood decreases during adolescence
- Estimation of GFR in pediatrics

$$GFR = \frac{kL}{SCr}$$

Age	k
Low birth weight < 1 year	0.33
Full term < 1 year	0.45
1–12 year	0.55
13–21 year (female)	0.55
13–21 year (male)	0.70

k = Proportionality constant

L = Length in cm

SCr = Serum creatinine in mg/dL

GFR = estimated glomerular filtration rate

(i.e., creatinine clearance) in mL/min/1.73 m²

SPECIFIC CONSIDERATIONS IN DRUG THERAPY

- Off-label Medication Use
 - Lack of pediatric safety information on 75% of drugs
 - Appropriate when no alternatives available
- **Routes of Administration and Drug Formulations**
 - (list routes of administration and when they are considered)
 - Compatibility and dosage reduction in parenteral administration
 - Solid vs liquid dosage forms of oral drugs
- **Common Errors in Pediatric Drug Therapy**
 - **Inappropriate off-label use, calculation errors, units of measurement**
 - **Careful review by pharmacists is important**

SPECIFIC CONSIDERATIONS IN DRUG THERAPY

- Complementary and Over-the-Counter Medication Use
 - CAM in pediatrics
 - OTC medications in pediatrics
- Medication Administration to Pediatric Patients and Caregiver Education
 - The issue of cooperation
 - Rounding vs exact dose administration
 - Points of caregiver education
- Accidental Ingestion in Pediatric Patients
 - Infants and young children at higher risk
 - Immediate measures to be taken...